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Scientists investigated the cell cycle in heart cells taken from mice 6 days before their birth and then at 4, 14 and 21 days after their birth.

Their results are shown in **Table 4**. Age 0 days = day of birth.

Table 4

Age / days	Percentage of heart cells undergoing mitosis	Percentage of heart cells undergoing DNA replication
–6	13.9	8.5
4	8.5	2.6
14	1.6	0.2
21	0.6	0.0

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Describe and explain the data in **Table 4**.

[2 marks]

[Extra space]

Describe how BrdU would be incorporated into new DNA during semi-conservative replication.

[illegible]

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Cells with BrdU in their DNA are detected using an anti-BrdU antibody with an enzyme attached.

Use your knowledge of the ELISA test to suggest and explain how the scientists identified the cells that have BrdU in their DNA.

[3 marks]

[illegible]

[Extra space]

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ANSWER IN THE SPACES PROVIDED**

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